



Study on influencing factors and forecast of global crude oil prices based on the hybrid model

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ABSTRACT

As one of the most important strategic resources in the world today, the influencing factors of crude oil have always been the focus of international attention. In this paper, not only the traditional factors were considered, but also the midstream and downstream products of crude oil factor was introduced, referring to the industry's viewpoint, when analyzing the influencing factors of global crude oil prices. Moreover, Least Absolute Shrinkage and Selection Operator (LASSO) model was used to filter and analyze these factors and Least Absolute Shrinkage and Selection Operator-Time Varying Parameter (LASSO-TVP) model was used to forecast global crude oil prices finally. The results showed that global crude oil prices were influenced by different influencing factors, and the same influencing factor had different effects on global crude oil prices in different time periods. Among them, midstream and downstream products of crude oil had a strong influence on global crude oil prices. It was further found that the LASSO-TVP model had the best predictive effect, indicating that the LASSO-TVP model could significantly improve the forecasting effectiveness of LASSO model and TVP model, and accurately forecast global crude oil prices.

1. Introduction

The plunge in global crude oil prices in the second half of 2014 hit oil-producing countries such as Venezuela and Russia hard, and even triggered regional war and social turmoil, which was described as a “black swan” event for global crude oil prices. Unlike previous global crude oil prices crashed, this one was caused not by a decrease in demand, but by an increase in supply [1]. Therefore, it was of great importance to identify the influencing factors of global crude oil prices and to forecast global crude oil prices for both crude oil importing and exporting countries [2].

Global crude oil prices forecasting faced significant challenges due to the complexity of the global crude oil prices formation mechanism [3,4]. Kilian [5] believed that demand factors were the main factors affecting global crude oil prices. Gallo et al. [6] examined the relationship between crude oil importing countries and Organization of Petroleum Exporting Countries (OPEC) by using an endogenous structure breakpoint test and found that supply factors were more correlated with global crude oil prices than demand factors. Therefore, supply factors represented by crude oil inventories and the Baltic Dirty Oil Index had a greater impact on global crude oil prices [7]. Abhishek Kumar and

Sushanta Mallick [8] believed that in the post-financial crisis period supply shocks had emerged as a source of short-run increases in oil prices, and demand shocks did not have a long-run effect on prices, unlike in the pre-financial crisis period. Ali Jadidzadeh and Apostolos Serletis [9] took monthly data (over the period from January 1976 to December 2012) and a structural VAR model to disentangle demand and supply shocks in the global crude oil market and investigate their effects on the real price of natural gas in the United States. The result showed that structural supply and demand shocks in the global crude oil market were correlated with natural gas.

In addition to supply and demand factors, financial factors were also considered to be an important factor affecting global crude oil prices [10,11]. Basher and Sadorsky [12] used a multi-factor risk analysis model to investigate the relationship between oil prices and emerging stock market returns, and the results showed that global crude oil prices fluctuations had a significant impact on emerging stock market returns. Zhang and Fan [13] estimated three possible spillover effects between international oil prices and the US dollar exchange rate, the results of which showed that there was an obvious spillover effect between them. After the 2008 financial crisis and the 2014 “black swan” event of global crude oil prices, Kilian et al. (2014) argued that speculative factors were

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also important factors affecting global crude oil prices, and then constructed a model to quantify speculative factors and studied the specific impact of speculative factors on global crude oil prices. Wang et al. [14] studied the relationship between factors (supply, demand, financial and speculative) and global crude oil prices by constructing a multi-granularity heterogeneous combination approach, indicating that supply factors and financial factors were negatively correlated with global crude oil prices and demand factors were positively correlated with global crude oil prices, and speculative factors played a role of “fueling the flames” in influencing global crude oil prices. The role of large movements was found to be stronger in the Bitcoin market than in the markets for crude oil and gold. As Bitcoin shared with these exhaustible resource commodities characteristics such as the fixed supply, the analysis of Bitcoin prices could generally learn from the analysis of exhaustible resource commodities. However, whereas the short-run supply of gold and oil were uncertain, there were no uncertainties on the Bitcoin supply-side [15]. Monge et al. [16] first investigated the statistical properties of the real oil prices as well as its log-transformation, along with the absolute and squared returns values. Then, they also addressed the following issue: Did the crude oil prices behave in the same way before and after a military conflict or geopolitical problem in the producer countries? The empirical results provided evidence of persistence and breaks in the oil prices series and stationary long memory in the absolute returns. The businesses research had shown that in addition to the supply and demand, financial and speculative factors, midstream and downstream products of crude oil, represented by fuel oil, could also affect crude oil supply of upstream in production chain through changes of social demand, and thus influenced changes of global crude oil prices [17].

Economic variables were often used for forecasting commodity prices, but technical indicators have received much less attention in the literature. Some paper demonstrated the predictability of commodity price changes using many technical indicators. Technical indicators were stronger predictors than economic indicators, and their forecasting performances were not affected by the problems of data mining or time changes [14].

In brief, midstream and downstream products of crude oil were introduced as a factor in this paper to examine its relationship with global crude oil prices, on the basis of summarizing the previous scholarly research and referring to the industry’s viewpoint.

Global crude oil prices were influenced by different factors at different time periods, and even the same influencing factor had different effects on global crude oil prices at different time periods [18]. Chai et al. [2] captured the trend change characteristics of global crude oil prices by constructing a time-varying Markov model, and screened out the influencing factors of global crude oil prices under different conditions through the Bayesian Model Averaging (BMA) model, verifying the hypothesis that the influencing factors of global crude oil prices were time-varying.

As there were varieties of influencing factors on global crude oil prices, the search cost of which was very high, however, too many factors could also lead to multicollinearity problems [19]. Therefore, it was necessary to screen out the factors that had a greater impact on global crude oil prices. LASSO model, because of its strong screening capability, had often been used in fields of selecting risk factors for bank loan defaults [20], and factors affecting mid-term power load [21], demonstrating better screening results yet. Moreover, scholars at home and abroad applied it to the energy field as well. Zhang et al. [22] constructed the LASSO model from the perspective of economics and statistics to screen out the most influential factors from the 34 influencing factors of global crude oil prices and forecast global crude oil prices, and achieved good results. Miao et al. [18] filtered data from the 26 influencing factors by LASSO model and stepwise regression model and forecasted the global crude oil prices, and the results showed that the LASSO model had stronger screening capabilities and better forecasting effects.

After screening out the factors that had a greater impact on global crude oil prices, it had been the academic focus of scholars’ attention that what kind of models should be used to predict the global crude oil prices [23]. Xu and Niu [24] introduced a novel model with the random vector functional link (RVFL) neural network were proposed. Then, both monthly and weekly West Texas Intermediate crude oil spot prices were used for multi-step-forward prediction. The experiment reports that: the proposed R-VMD-RVFL (Rolling-Variational Mode Decomposition-Random Vector Functional Link) model achieved superior and robust accuracy in both monthly and weekly crude oil price predictions, especially when the forecast horizon was relatively long. Guo et al. [25] introduced artificial intelligence methods to evaluate the optimal forecasting strategy for China’s crude oil futures prices. They used machine learning to process historical information, volatility and non-linear features. Using daily data from March 26, 2018 to February 28, 2023, they estimated the forecasting effects of gate recurrent unit (GRU), artificial neural networks (ANN), recurrent neural networks (RNN), support vector machines (SVM), multilayer perceptron (MLP), and long short-term memory (LSTM) models on China crude oil futures, respectively. With a series of evaluation tests, which demonstrated that the gate recurrent unit (GRU) model outperformed other models in terms of forecast accuracy and performance for China’s crude oil futures prices.

Li et al. [26] proposed a new, novel crude oil price forecasting method based on online media text mining, with the aim of capturing the more immediate market antecedents of price fluctuations, and their empirical results suggested that the proposed topic-sentiment synthesis forecasting models performed better than the older benchmark models. In addition, text features and financial features were shown to be complementary in producing more accurate crude oil price forecasts. Wu et al. [27] took a novel model based on ensemble empirical mode decomposition (EEMD) and long short-term memory (LSTM) to forecast the global crude oil prices, empirical results demonstrated that the proposed model still work well when the number of decomposition results varies. four methods of the least square support vector machine (LSSVM), genetic programming (GP), artificial neural network (ANN), and auto-regressive integrated moving average (ARIMA) were initially used to forecast monthly oil price (MOP), daily gas price (DGP), and annual interest rate (AIR). Zhao et al. [28] introduced financial market factors and crude oil news as forecasters, and proposed a novel hybrid model with two-layer multivariate decomposition. The results suggested that the second multivariate decomposition for the high-frequency sub-component could significantly improve the forecasting accuracy, and the forecasting performance of the proposed model outperformed all the benchmark models. Xing and Zhang [29] explored different specifications of shrinkage methods to forecast global crude oil prices, and examined the forecasting performances from both statistical and economic perspectives, and they considered two popular penalties and Huber loss function in penalized regressions. The results indicated that the out-of-sample performance of different shrinkage methods depend on the forecasting horizons.

Compared with traditional models, although machine learning method had a certain advantage in solving non-linear problems, it was prone to making over-fitting and local minima mistakes [30].

Since both traditional econometric economics models and machine learning approaches assumed that the coefficient of global crude oil prices and its influencing factors was fixed, which means that the relationship between global crude oil prices and influencing factors was constant within the sample [21,31]. However, their relationship changed over time, in fact, especially after significant changes in global crude oil prices. In contrast, the TVP model assumed that the relationship mentioned above was time-varying and could overcome the problem of fixed coefficient. Consequently, the TVP model was introduced in this paper to capture the time-varying relationship between global crude oil prices and its influencing factors and to forecast global crude oil prices in conjunction with the influencing factors. What’s more, we

introduced the TVP model, first used to capture the time-varying relationship between tourism time series and its influencing factors [32], to forecast global crude oil prices for the very first time.

Summing up the above, the midstream and downstream products of crude oil could influence changes of global crude oil prices through its own price changes, and thus made a significant impact on global crude oil prices. As a consequence, we introduced midstream and downstream products of crude oil as an influencing factor of global crude oil prices for the first time, based on the four categories of influencing factors, namely supply, demand, finance and alternative products. As there were noise variables and multicollinearity problems among these influencing factors, we used LASSO model to solve the multicollinearity problem among these influencing factors and screens and analyzes them, and then we put the influencing factors screened by LASSO model into TVP model to construct the LASSO-TVP model to forecast global crude oil prices.

The contribution of this paper were mainly as follows: Firstly, we complemented the research gap on the relationship between the midstream and downstream products of crude oil and global crude oil prices by domestic scholars; secondly, it used a new parameter estimation method to estimate the parameters of the LASSO model, overcoming the parameter instability problem of the cross-validation method in this paper; finally, the time-varying relationship between the global crude oil prices and its influencing factors was well analyzed, by studying on the influencing factors of the rolling screening of LASSO model, and the forecasting accuracy of the global crude oil prices was also improved through the LASSO-TVP model.

The remainder of the paper is organized as follows. Section 2 reviews the relevant models of crude oil forecasting. Section 3 is selection and analysis of the influencing factors of global crude oil prices. Section 4 presents the empirical results, followed by the conclusions in section 5.

2. Construction of the LASSO-TVP model

2.1. LASSO model

Many scholars had constructed linear regression models to study the relationship between global crude oil prices and those influencing factors. The optimal objective of the ordinary linear regression model was to minimise the residual sum of squares of the objective function, which could be expressed as follows:

$$Y = \alpha + \beta X + \varepsilon \quad (1)$$

$$G(\beta) = \sum_{i=1}^N \left(y_i - \alpha_i - \sum_{j=1}^P x_{ij} \beta_j \right)^2 \quad (2)$$

In eq. (1) and eq. (2), $Y = (y_1, y_2, \dots, y_i)$ indicated a dependent variable matrix, y_i was the dependent variable, $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_i)$ indicated a intercept matrix, α_i was the intercept, $X = (x_{i1}, x_{i2}, \dots, x_{ij})$ indicated the independent variable, in which x_{ij} was a independent variable, and $\beta = (\beta_1, \beta_2, \dots, \beta_j)$ indicated a coefficient matrix, in which β_j was the coefficient, N was the array value, and P was the number of independent variables.

When the number of independent variables was large or there was a multicollinearity problem among the independent variables, β_i in eq. (2) could not be estimated, and ordinary linear regression failed. However, there were so many factors that affected global crude oil prices, with different functions, and ordinary linear regression could not be used, because of the noise variables and multicollinearity problems between variables, so these influencing factors must be identified and screened.

At present, the main methods to solve such problems were stepwise regression and regularization methods [20]. The stepwise regression method came into being earlier, which was simpler and easier to operate, but the parameters were difficult to estimate, because this method

would lead to a geometric increase in computation with the increase of variable dimensionality. In contrast, the regularization method was the most commonly used method currently, and the principle of this method was to add a regularization term (constraint function related to the coefficient) on the basis of the loss function to constrain the complexity of the model. The optimal objective function with the regular term was:

$$G(\beta) = \sum_{i=1}^N \left(y_i - \alpha_i - \sum_{j=1}^P x_{ij} \beta_j \right)^2 + \lambda f(\beta) \quad (3)$$

The LASSO algorithm was the most representative method of regularization methods, proposed by Ref. [33], and this method could set the coefficients of less influential factors to zero and solved the problem of multicollinearity between variables as well, widely used in the field of variable selection. The optimal target function of the LASSO algorithm was:

$$G(\beta) = \frac{1}{N} \sum_{i=1}^N \left(y_i - \alpha_i - \sum_{j=1}^P x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^P |\beta_j| \quad (4)$$

In this paper, LASSO algorithm was introduced to forecast global crude oil prices. In eq. (4), y_i denoted the log value of global crude oil prices in month i , x_{ij} was the log value of the influencing factor j of global crude oil prices in month i , and β_j the shrinkage estimation value of regression coefficients in the LASSO algorithm, estimated from the data before month i . And α_i was the intercept item, λ was the non-negative regularized parameter of β . The larger the value was, the greater the constraint on coefficient was, and the more terms may be zero.

2.2. TVP model

To examine the extent to which parameters in crude oil forecasting models changed over time, recursive OLS methods had been used for long-term static models. Although the recursive OLS approach was very effective for structurally stable models, it was indistinguishable from traditional OLS regression when all data points were used to estimate the model. Therefore, the recursive OLS model cannot adapt to structural changes when performing out-of-sample predictions.

The TVP model allowed parameters to vary over time, able to overcome the drawbacks of recursive OLS regression methods, so it was more effective in dealing with structural changes in econometric models. In addition, the TVP model could constantly update the estimation process and gave the latest information more weight during the process to capture the structural breakpoints of global crude oil prices in time. The equation of the TVP model was as follows:

$$y_t = \alpha + \sum_{j=1}^k \beta_{jt} x_{jt} + \sum_{i=1}^p \phi_{it} y_{t-i} + \varepsilon_t \quad (5)$$

$$\beta_{jt} = \varphi_{j0} + \sum_{h=1}^m \varphi_{jh} \beta_{j,t-h} + \omega_{jt} (j=0, 1, 2, \dots, k), \omega_{jt} \sim N(0, \sigma_\omega^2) \quad (6)$$

$$\phi_{it} = \varphi_{i0} + \sum_{k=1}^m \varphi_{ik} \phi_{i,t-k} + \gamma_{it} (i=1, 2, \dots, N), \gamma_{it} \sim N(0, \sigma_\gamma^2) \quad (7)$$

Equation (5) denoted the entire system equation, in which y_t indicated the log value of global crude oil prices in month t , x_{jt} denoted the log value of influencing factor j in month t , β_{jt} was the coefficients of influencing factor j in month t , y_{t-i} denoted the log value of the i th order lag of global crude oil prices, ϕ_{it} denoted the coefficient of lag term of global crude oil prices, α denoted the intercept item, and ε_t denoted the disturbance item.

Equation (6) and equation (7) denoted the transition equations used to estimate how parameters of the system equation change over time, usually defined as an autoregressive process, and sometimes, other

exogenous variables will be added to the equation, but they were estimated in the same way. In eq. (6), β_{jt} denoted the coefficient of the influencing factor j in month t , β_{jt-h} denoted the coefficient of the influencing factor j from month t to month h , φ_{jh} denoted the coefficient of β_{jt-h} , namely the weight, and ω_{jt} denoted disturbance term. It could be seen from eq. (6) that the coefficient of influencing factors at moment t was equal to the weight sum plus disturbance term of coefficients lagging 1 to m , that was, the coefficient of current period was determined by the weight sum and disturbance term of the coefficients of the previous periods. The most common form of transition equation was stochastic process, as was shown below:

$$\beta_{jt} = \beta_{jt-1} + \omega_{jt} \quad (8)$$

$$\phi_{it} = \phi_{it-1} + \gamma_{it} \quad (9)$$

Equation (8) and equation (9) shared the same estimation approach. From eq. (8), it could be seen that β_{jt} was equal to sum of the previous coefficient β_{jt-1} plus the disturbance term ω_{jt} , indicating that the equation assigned a weight of 100 % to the coefficient of the previous period and a weight of 0 to the coefficients of the remaining lagged factors, namely, it assigned the maximum weight to the latest information. Moreover, the model parameters were all estimated via the method of Kalman filtering.

2.3. LASSO-TVP model construction and solution

Through Section 2.1 and 2.2, although the Lasso and TVP models had certain advantages in forecasting global crude oil prices, there were still some problems yet. Among them, the LASSO model could screen out strong influence factors, but it could not accurately predict the nonlinear series with high volatility; the TVP model could capture the structural breakpoint of global crude oil prices by updating the parameters in a timely manner, but its prediction accuracy and operational efficiency could be disturbed due to the large number of influencing factors of global crude oil prices and the model's inability to identify and screen them. In view of the multicollinear problem among the influencing factors in crude oil prices forecasting and the nonlinear problem of crude oil price sequence itself, we combined the LASSO model and TVP model together to construct the LASSO-TVP model to screen the influencing factors of global crude oil prices and forecast the global crude oil prices. The specific process was as follows:

Firstly, the influencing factors of global crude oil prices were screened by the LASSO model, and the screened influencing factors with a frequency of more than 10 % were collated and analyzed. The LASSO model screened out the influencing factors by minimizing the optimal target function of eq. (4), so that the estimated value of some parameters was compressed to zero. The Alternating Direction Method of Multipliers (ADMM) method was used in this paper for parameter estimation, which took the following specific form:

1. Simplify eq. (4) into the following form:

$$J(\beta) = ESS(\beta) + \lambda l_1(\beta) \quad (10)$$

In eq. (10), $ESS(\beta)$ represented the sum of squared errors and $\lambda l_1(\beta)$ denoted the constraint term. Since the constraint term became an absolute value, it was not derivable at the zero point, so the coordinate descent method was adopted here. This was an iterative non-gradient optimal method in which the search was performed along only one direction in each iteration step, while other parameters were fixed, and the objective function converged to a minimum value by cycling in different coordinate directions. The specific solution was as follows:

2. Found the partial derivative for each β_j in $ESS(\beta)$ and $\lambda l_1(\beta)$, and made the sum of the derivative functions zero, so as to obtain the optimum $\hat{\beta}$.

$$\frac{\partial ESS(\beta)}{\partial \beta_j} + \frac{\partial \lambda l_1(\beta)}{\partial \beta_j} = \begin{cases} -2m_j + 2\beta_j n_j + \lambda = 0 \\ [-2m_j - \lambda, -2m_j + \lambda] = 0 \\ -2m_j + 2\beta_j n_j - \lambda = 0 \end{cases} = 0 \quad (11)$$

$$\Rightarrow \beta_j = \begin{cases} \left(m_j - \frac{\lambda}{2}\right) / n_j, m_j > \frac{\lambda}{2} \\ 0, m_j \in \left[-\frac{\lambda}{2}, \frac{\lambda}{2}\right] \\ \left(m_j + \frac{\lambda}{2}\right) / n_j, m_j < -\frac{\lambda}{2} \end{cases}$$

Where $m_j = \sum_{i=1}^N x_{ij} (y_i - \sum_{k \neq j} \beta_k x_{ik})$, $n_j = \sum_{i=1}^N x_{ij}^2$.

3. The λ value was generally determined by the cross-validation method, but the sample set was randomly divided into training sets and verification sets in this way, which would not only disturb the order of time sequence, but also made the regression results unstable. In this paper, a new parameter estimation method was used, which could strictly divide the sample set according to the time sequence and could fix the training sets and validation sets, making the regression results more stable and thus overcoming shortcomings of the cross-validation method [34]. The specific steps were as follows:

- ① Given an initial set of λ , said $\lambda^{(1)}$, we further divided the estimation period (i.e., the period from month 1 to t) into an “in-sample” estimation period and an “out-of-sample” evaluation period.
- ② We estimated the regression coefficients $\hat{\beta}$ of the LASSO by solving the optimization problem of Eq. (10), respectively, using the “in-sample” data from month 1 to j ($j < t$) with $\lambda = \lambda^{(1)}$. The coefficient estimation used in this step was based on the ADMM algorithm (Boyd et al., 2011).
- ③ We used the estimated coefficients $\hat{\beta}$ to produce “out-of-sample” forecasts from month $j + 1$ to t and then computed the mean squared prediction error (MSPE) based on these forecasts, which was conditional on $\lambda^{(1)}$.
- ④ We repeated steps 1 to 3 with different values of λ , say $\lambda^2, \lambda^3, \dots, \lambda^n$, selected the optimal value of λ that provided the lowest value of MSPE.
- ⑤ $\hat{\lambda}$ Given, we implemented the ADMM algorithm to estimate the regression coefficients $\hat{\beta}$ using all the data over the entire estimation period from month 1 to t , instead of only its “in-sample” part from month 1 to j .

Secondly, eq. (11) showed that the coefficient of the influencing factors estimated by the LASSO model were ($\beta_1 = k_1, \beta_2 = 0, \beta_3 = 0, \dots, \beta_j = k_j$) and the influencing factors with a non-zero coefficient were recorded as $(x_1, x_j, \dots, x_n)$. The LASSO model was used to conduct a fixed-window rolling screening, and the influencing factors with non-zero coefficients in the screening results were statistically analyzed to identify the influencing factors with a frequency of more than 10 %, which were recorded as $(x_1, x_j, \dots, x_k)$.

Thirdly, the LASSO-TVP model was constructed by substituting $(x_1, x_j, \dots, x_k)$ into the TVP model. Since we used only influencing factors in forecasting global crude oil prices, with no lag value of global crude oil prices included. Therefore, the form of the TVP model constructed in this paper was as follows:

$$y_t = \alpha + \sum_{j=1}^k \beta_{jt} x_{jt} + \varepsilon_t \quad (12)$$

$$\beta_{jt} = \beta_{jt-1} + \omega_{jt} \quad (13)$$

y_t denoted the logarithmic value of global crude oil prices, and $(x_1, x_j, \dots, x_k)$ indicated the influencing factors with a frequency of exceeding 10 % as screened by LASSO.

Fourthly, estimated the prediction result of LASSO-TVP model and then calculated its error value.

2.4. No-change model

Baumeister and Kilian [35] argued that crude oil spot prices were stochastic wandering processes and that change in global crude oil prices was unpredictable. Therefore, the best forecast for crude oil spot prices was the current spot prices:

$$\hat{p}_{t+h} = p_t \quad (14)$$

In order to further validate the forecasting effectiveness of the four models, in particular the LASSO-TVP model, for global crude oil prices, we introduced two widely used statistical error indicators, namely the mean absolute percentage error (MAPE) representing relative values and the mean absolute error (MAE) representing absolute values. The indicators were assessed as follows:

$$MAE = \frac{1}{T-N} \sum_{t=N+1}^T |y_t - \hat{y}_t| \quad (15)$$

$$MAPE = \frac{1}{T-N} \sum_{t=N+1}^T \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100\% \quad (16)$$

y_t and $\hat{y}_t$ represented the actual prices and predicted prices of crude oil, respectively, and T and N represented the full-sample observation value and the in-sample observation value, respectively.

Next, we utilized the Diebold-Mariano (DM) statistic to compare the prediction errors of the two models.

$$DM = \frac{\bar{d}}{\sqrt{Var(\bar{d})}} \quad (17)$$

Where $\bar{d} = \frac{1}{n} \sum_{t=1}^n d_t$, $d_t = (y_t - y_{1,t})^2 - (y_t - y_{2,t})^2$, $Var(\bar{d}) = \frac{1}{n} (\gamma_0 + 2 \sum_{k=1}^n \gamma_k)$ and $\gamma_k = Cov(d_t, d_{t-k})$. $y_{1,t}$ represented the predicted values obtained by the first model, whereas $y_{2,t}$ obtained by the second model. The only assumption of DM statistic was that the two models had an equal number of predictions.

3. Selection and analysis of the influencing factors of global crude oil prices

Crude oil had not only commodity attributes, but also strong

financial attributes. The former was mainly reflected in the fact that global crude oil prices were influenced by supply and demand factors, while the latter was mainly reflected in the fact that global crude oil prices were influenced by financial factors such as the (United States) US dollar index, the Volatility Index (VIX) and interest rates. Due to the uneven distribution of crude oil resources and the huge differences in the consumption demand for crude oil among countries all over the world (see Table 1), the supply and demand factors for crude oil have become the most fundamental influencing factors on global crude oil prices.

With the emergence of the ‘petrodollar’ and the development of national financial markets, crude oil had become more speculative as a financial asset, therefore financial factors had become important in influencing the price of crude oil gradually [22]. We would analyze 33 influencing factors such as the supply, demand, financial, alternative products, and midstream and downstream products of crude oil.

- (i) Supply factors. Supply factors mainly included OPEC spare capacity, US shale oil production, Energy Information Administration (EIA) crude oil inventories and US crude oil production, among which US shale oil production had the greatest impact on global crude oil prices [36]. Compared with conventional crude oil, shale oil was more resilient and price elastic. As a result, OPEC’s choice to increase production at the beginning of the shale oil revolution did not result in a reduction in shale oil production. When shale oil production continued to grow, OPEC used production cuts to support and stabilize oil prices, yet rising global crude oil prices further boosted shale oil production. Thus, it will be a cycle for a long time to come: conventional gasoline production was reduced to support and stabilize oil prices, unconventional oil and gas resources reached a tipping point to increase production and suppress prices, and conventional oil and gas production was then reduced to raise prices again (see Fig. 1).
- (ii) Demand factors. The demand factor mainly includes the economic growth rate of China and the United States, the PMI index and the RBOB gasoline - WTI spot price spread, etc., which mainly affected global crude oil prices through crude oil consumption [37]. And PMI index was the most representative among them, which could reflect the growth and recession of the manufacturing and service industries, and the development of the economy to a certain extent as well, and had a certain influence on the demand of global crude oil prices [38,39]. What’s more, the volatility of the PMI index was higher than that of global crude oil prices.
- (iii) Financial factors. The financial attributes of crude oil were mainly reflected in its influence of financial factors such as US dollar liquidity, US dollar index, US bond real yields, VIX index and credit risk premium, among which US dollar liquidity, US dollar index and credit risk premium could all reflect the value of the US dollar and shared the same mechanism [40]. Therefore, this paper took the US dollar index as an example. US dollar index. The level of the US dollar index directly reflected the value

Table 1
Crude oil production and consumption.

	North America	South America	Europe	Middle East	Africa	Asia Pacific	OECD(Organization for Economic Co-operation and Development)	Non-OECD
The proportion of Crude oil production accounting for the world crude oil	20.5 %	9.3 %	19.8 %	31.7 %	9.3 %	9.4 %	24.6 %	75.4 %
The proportion of Crude oil consumption accounting for the world crude oil consumption	24.3 %	7.8 %	20.4 %	9.3 %	4.3 %	33.9 %	48.3 %	51.7 %
The proportion of crude oil consumption accounting for production	1.2	0.9	1.1	0.3	0.5	3.7	2.0	0.7

Data Source:NYMEX official website

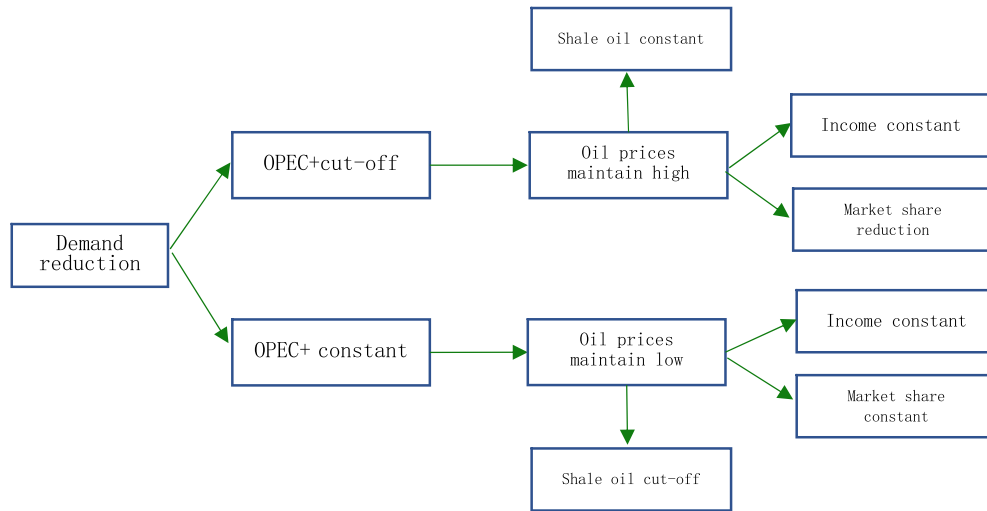


Fig. 1. Shale oil production and OPEC production.

of the US dollar. Since international crude oil was denominated in US dollars, the trend of international oil prices was opposite to that of the U.S. dollar index. Namely, when the US dollar index rose, the price of crude oil denominated in US dollars will fall; and vice versa. The VIX Index. The financial attributes of crude oil were also reflected in its exposure to market risk appetite as a risk asset. We introduced the VIX as a measure of market risk appetite in this paper. Considering that the VIX index, which measured the degree of market panic, was inversely correlated with the degree of risk appetite, the price of crude oil as a risk asset generally moved in the opposite direction to the VIX.

- (iv) Midstream and downstream products of crude oil. The demand for gasoline, paraffin, diesel, naphtha, etc., midstream and downstream products of crude oil, would fluctuate with the increase or decrease in global crude oil prices, because the price of crude oil directly affected the cost of midstream and downstream products of crude oil, and the consumption and price of midstream and downstream products of crude oil would also affect global crude oil prices [41]. In general, the price of crude oil was positively correlated with the price of its midstream and downstream products [42].

4. Empirical analysis

4.1. Sample selection and data source

Both Brent global crude oil prices and WTI global crude oil prices were representative of international global crude oil prices and were widely used in international crude oil pricing. Although WTI was one of the most influential crude oil in the world, and WTI futures contracts maintained a dominant role in price discovery relative to Brent [43], it was used more for the US's own consumption and was highly localized, and therefore had a smaller share of the world crude oil export market [41,44]. However, in contrast, the majority of crude oil exports to Europe (Beihai, Middle East, Latin America and Canada) were denominated in Brent crude oil, which accounted for over 65 % of the crude oil spot market, at the same time, more than 85 % of China's imported crude oil was based on Brent crude oil prices. Therefore, we used Brent crude oil as a proxy for international global crude oil prices.

Since the OPEC spare capacity, US shale oil production and Chinese crude oil consumption among the influencing factors of Brent crude oil prices were highly volatile [37], our model took logarithms of Brent crude oil prices and all its influencing factors in Table 2 to reduce their volatility. At the same time, as the influencing factors such as China's

Table 2
Influencing factors of global crude oil prices.

分类	variable	source
supply	OPEC residual capacity	EIA
	US(United States) shale oil production	EIA
	EIA crude oil inventory	EIA
	American crude oil production	EIA
	Global crude oil supply	EIA
	Number of global rigs	EIA
demand	Number of American rigs	EIA
	Chinese economic growth	WIND
	US PMI(Purchasing Managers' Index)	WIND
	China PMI	WIND
	RBOB-WTI(Reformulated Gasoline Blendstock for Oxygen Blending-West Texas Intermediate) spot price spread	EIA
	American refining production	WIND
	US inventory	WIND
	American crude oil export volume	WIND
	Chinese crude oil import and processing volume	WIND
	Chinese crude oil consumption	WIND
finance	M2/GDP(Broad money/Gross Domestic Product)	WIND
	LIBOR(London Interbank Offered Rate)	WIND
	US dollar index	WIND
	US bond real yield	WIND
	Inflation rate	WIND
	VIX(Volatility Index)	WIND
	Credit risk premium	WIND
	Gold price	WIND
	WTI futures position	WIND
	CRB(Commodity Research Bureau) index	WIND
Replacement	S&P 500(Standard & Poor's 500) Index	WIND
	HH(Henry Hub) Natural gas price	WIND
	Coal consumption	WIND
midstream and downstream products of crude oil	Gasoline ¥/ton	EIA
	diesel fuel ¥/ton	EIA
	Kerosene ¥/ton	EIA
	Naphtha ¥/ton	EIA

PMI index and US crude oil exports were only available on a monthly basis, all the data frequencies selected in this paper were monthly and the sample time period selected was from January 2005 to June 2023.

4.2. Test for multicollinearity of influencing factors

Since the premise of LASSO model was that there was multicollinearity among influencing factors, we first tested whether there was

multicollinearity among variables and judge whether variables could be eliminated by the LASSO model.

As shown in Table 3, the mean VIF of the influencing factors was 106.68, which was much greater than 10. In addition, the VIF of 25 influencing factors was greater than 10, accounting for 76 % of the total. Therefore, there was a serious multicollinearity among the influencing factors. The VIF of US shale oil production was the largest, which was 1110.16, and the multicollinearity was the strongest. This was because US shale oil had a greater impact on the supply, demand, alternative products and other factors of crude oil, and had a strong marginal utility. The second was Chinese crude oil consumption, whose VIF was 1085.49, which also had a strong multicollinearity. This was because China was the world's largest importer of crude oil, with a large demand for crude oil and a high dependence on foreign countries, so it was related to many influencing factors of crude oil.

The VIF of Coal consumption, S&P 500 Index, Gasoline ¥/ton, Number of global rigs, Chinese economic growth, etc were greater than 10, which showed that there were multicollinearity in five types of factors, namely supply, demand, finance, alternative products and midstream and downstream products of crude oil. Since the LASSO model could eliminate multicollinearity, these influencing factors could be screened using the LASSO model.

4.3. Time-varying relationship between influencing factors and global crude oil prices

Sample time period from January 2005 to January 2014 was used as in-sample data and from February 2014 to June 2023 as out-of-sample data. The LASSO model was introduced for one-step rolling screening of global crude oil prices and their influencing factors in this paper, with the length of in-sample data as a fixed window and a total rolling length of 113 months. Since the LASSO model screened out different influencing factors when rolling to different time periods, this paper would focus on analyzing the reasons for the changes in the influencing factors screened out by the LASSO model. We could know from Section 4.2 that these influencing factors had serious multicollinearity problems, so LASSO model could be used.

For the influencing factors were divided into five categories, each with a different number. Therefore, the percentage of selected influencing factors (the percentage of selected factors in such factors) was taken to study the time-varying relationship between influencing factors and global crude oil prices [45]. Due to the fact that the out-of-sample

Table 3
Test for multicollinearity of influencing factors.

Variable	VIF	Variable	VIF
US shale oil production	1110.16	Naphtha ¥/ton	32.03
Chinese crude oil consumption	1085.49	Kerosene ¥/ton	31.45
Coal consumption	136.02	EIA crude oil inventory	30.06
S&P 500 Index	134.24	US dollar index	25.84
Gasoline ¥/ton	95.80	RBOB-WTI spot price spread	19.43
Number of global rigs	92.18	Credit risk premium	14.19
Number of American rigs	91.86	OPEC residual capacity	13.09
diesel fuel ¥/ton	85.92	US PMI	12.32
Gold price	82.15	HH Natural gas price	11.00
M2/GDP	62.91	Chinese crude oil import and processing volume	9.10
American crude oil production	56.76	VIX	7.31
US inventory	50.90	American refining production	7.28
Chinese economic growth	45.90	Inflation rate	6.50
CRB index	42.62	China PMI	5.23
LIBOR	41.40	US bond real yield	4.63
Global crude oil supply	37.48	WTI futures position	4.16
American crude oil export volume	35.12		
Mean VIF	106.68		

rolling cycle was relatively long, quarters were used as the unit for analysis. This section began with the analysis of the total influencing factors selected by LASSO and then analyzed the influencing factors in detail by dividing them into four categories: supply, demand, finance and midstream and downstream products of crude oil. The group with the higher percentage selected was the main cause of crude oil prices fluctuation over a period.

4.3.1. The total time-varying relationship between influencing factors and global crude oil prices

As Fig. 2 displayed, global crude oil prices sharply fluctuated and more influences were selected during the four periods from the first quarter of 2014 (2014Q1(Quarter1)) to the fourth quarter of 2014 (2014Q4), from the third quarter of 2017 (2017Q3) to the fourth quarter of 2019 (2019Q4) [46], from the first quarter of 2020 (2020Q1) to the third quarter of 2020 (2020Q3), and from the first quarter of 2022 (2022Q1) to the third quarter of 2022 (2022Q3). For example, global crude oil prices suddenly plummeted from US\$115/barrel in the middle of the year to US\$61/barrel at the end of the year, and the percentage of selected influencing factors increased from 22 % to 48 % in the second half of 2014 [47].

Global crude oil prices were relatively stable, with fewer influencing factors selected, and the percentage was only between 5 % and 15 %, between the first quarter of 2015 (2015Q1) and the second quarter of 2017 (2017Q2).

However, from the second half of 2017, global crude oil prices fluctuated dramatically due to the impact of the US dollar interest rate hike and the US-China trade war, which first rapidly increased from \$46.37/barrel to \$78.36/barrel, falling to \$57.36/barrel within a short period, and then subsequently recovered to \$72/barrel, fluctuating sharply [48]. More influencing factors were selected at this time, with their percentages all above 30 %, and with a maximum of 55 %. This suggested that influencing factors that caused violent fluctuations in global crude oil prices were complicated. Therefore, this paper provided specific analysis from four perspectives: supply, demand, finance and midstream and downstream products of crude oil.

The Corona Virus Disease 2019 (COVID-19) in 2020 had hit global crude oil prices heavy, the global economy had slowed down, air transportation, rail transportation, travel, etc., and the demand for crude oil had plummeted. Global crude oil prices had plummeted, which were from \$67.31/barrel in 2019Q4 to \$14.85/barrel in 2020Q1, a plunge of 80 %, WTI crude oil futures once fell to -\$37/barrel, and crude oil prices were hit hard. The combination of midstream and downstream products of crude oil, demand factors, and financial factors lead to the plunge of global crude oil prices, therefore, the percentage of influential factors selected increased from 29 % to 47 %.

The Russia-Ukraine war in 2022 caused global crude oil prices to skyrocket from \$77.24/barrel in 2021Q4 to \$119.78/barrel in 2022Q2, an increase of 55 %. The combination of oil supply and financial factors were the main cause, so the percentage of influential factors selected has increased from 16 % to 30 %.

4.3.2. Time-varying relationships between supply, demand, finance and crude oil midstream and downstream products and global crude oil prices

4.3.2.1. Time-varying relationship between supply factors and global crude oil prices. As shown in Fig. 3, global crude oil prices plummeted in the second half of 2014, the percentage of selected supply factors increased sharply, indicating that the main cause of this global crude oil prices collapse was an increase in crude oil supply, as analyzed specifically below:

- (i) The U.S. shale oil production had been increasing significantly. With the completion of the US Seaway oil pipeline, the daily output of shale oil in the United States reached 8.4 million barrels

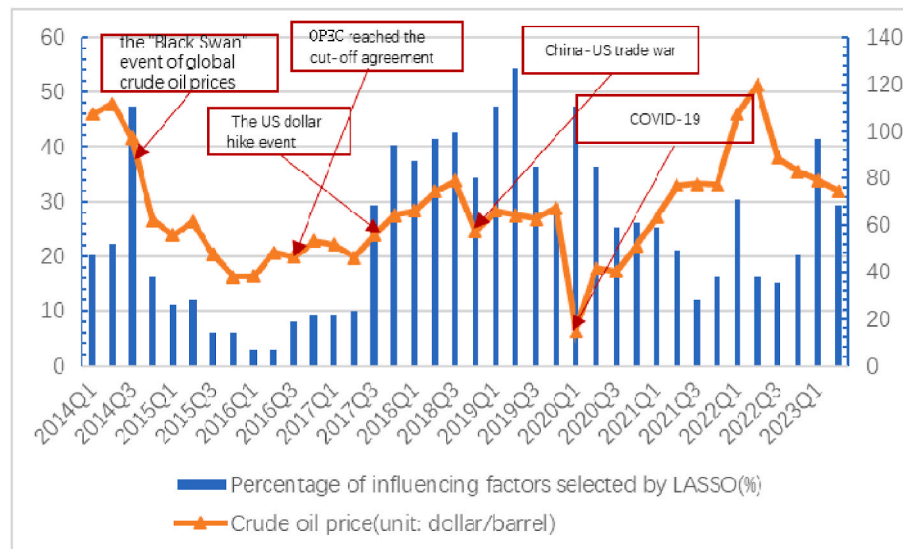


Fig. 2. The percentage of the total influencing factors selected by LASSO (left axis) and global crude oil prices (right axis).

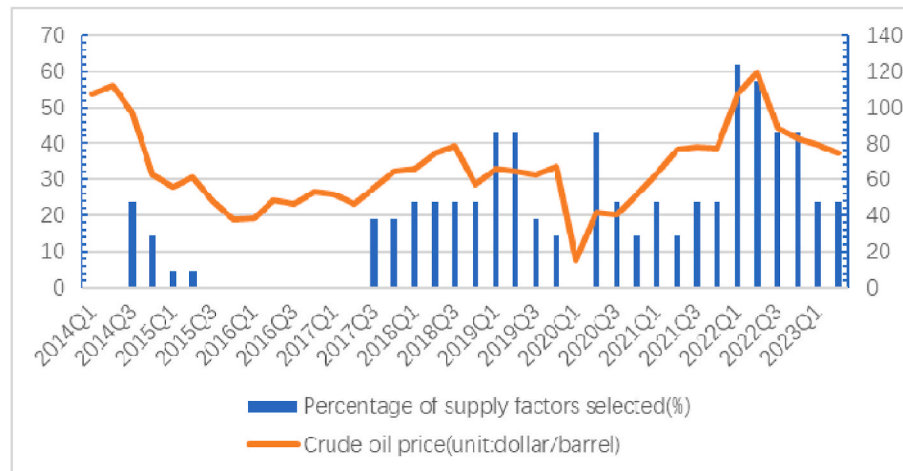


Fig. 3. The percentage of the supply factors selected by LASSO(left axis) and global crude oil prices (right axis).

per day, an increase of 1.5 million barrels per day from the average daily production in 2013, and it rose to 9.5 million barrels per day in March 2015, further reducing the United States' demand for external crude oil, and gradually eroding the market share of other crude oil exporting countries, especially OPEC.

- (ii) Traditional oil producing countries such as Iraq and Libya were gradually recovering from the war. In June 2014, Iraq's crude oil exports have increased to 2.53 million barrels per day, while Libyan's crude oil production had also increased to 925,000 barrels per day, and Saudi Arabia's also increased its crude oil production to 10 million barrels per day, further increasing the international crude oil supply.
- (iii) The OPEC failed to reach an agreement on production cut-off agreement. On November 27, 2014, a new round of OPEC meetings was held to limit the US shale oil revolution, and it was decided to continue to increase crude oil supply and limit exports of US shale oil. As a result, OPEC crude oil production reached 30.3 million barrels per day, which was already above the daily production limit.

In contrast, during the two years of 2015 and 2016, global crude oil

prices remained low and the trend was relatively stable, with less change in international crude oil supply and demand, and therefore the supply factors had less impact and was less likely to be selected. And from the second half of 2017, global crude oil prices rebounded rapidly, rising from \$49.5 per barrel in June 2017 to \$78.9 per barrel in September 2018, mainly due to two supply factors, namely the agreement on production cut-off reached by the world's major crude oil exporters and the reduction in world crude oil inventories. The specific analysis was as follows:

- (i) The OPEC reached a new agreement on production cut-off. On November 30, 2016, the OPEC reached its first agreement on production cut-off in eight years, deciding to cut production by 1.2 million barrels per day from January 1, 2017, followed by the non-OPEC members such as Russia, Kazakhstan, Venezuela and Mexico joining and cutting production by 558,000 barrels/day. Then the international crude oil supply was significantly reduced and global crude oil prices continued to strengthen.
- (ii) World crude oil inventory had been continued to decrease. World crude oil inventories continued to fall as a result of the reduction in crude oil supply due to OPEC's production cut-off, with world crude oil inventories standing at 2.966 billion barrels at the end

of 2016 and falling to 2.964 billion barrels at the end of 2017. US commercial crude oil stocks also fell to 457 million barrels, falling 14.64 % year-on-year, due to the peak consumption season.

The world global crude oil prices had been maintained at a higher level of around US\$70 per barrel since 2018, due to the effect of the production cut-off agreement and the continued reduction in crude oil inventories, resulting in a shortage of crude oil supply, and the supply factor having a greater impact.

The COVID-19 in 2020 had a major impact on the international energy market, and global crude oil prices had plummeted due to a sharp decline in crude oil demand, with WTI crude oil futures prices even falling below -\$37/barrel at one point. In order to reduce losses and boost oil prices, in April 2020, OPEC + reached the largest cut-off production agreement in history, initially reducing production by 9.7 million barrels per day, and since then, it had maintained a high cut-off production rate, and effectively adjusted the market supply and demand balance through additional production cuts by some countries and accelerating the frequency of policy adjustments. At the end of 2021, world oil supply was 95.4 million barrels per day, 1.5 million barrels per day less than in 2020. Driven by the OPEC + cut-off production agreement, global crude oil prices recovered rapidly, which was from \$14.85/barrel in the first quarter of 2020 to \$51.22/barrel in the fourth quarter. Due to the joint production reduction of more than a dozen members of OPEC+, the supply reduction was the reason for the rebound of global crude oil prices from the second quarter of 2020, so the probability of supply factors being selected in the second quarter of 2020 was 42.86 %, and the later two quarters of 2020 have been selected.

On February 24, 2022, the Russia-Ukraine war broke out, the international crude oil supply decreased, and global crude oil prices skyrocketed. In less than half a year, the global crude oil prices rose from \$77.24/barrel to \$119.78/barrel. According to the «BP Statistical Review of World Energy», Russia's oil reserves accounted for about 6.2 % of the reserves of the world, ranking sixth in the world, of which crude oil and condensate production ranked second in the world. The sanctions imposed by Europe and the United States on Russia seriously affect the world crude oil supply, and the reduction of supply was the root cause of the increases of global crude oil prices. As a result, the probability of the supply factors selected in 2022 had always been more than 40 %, with a maximum of 61.9 %.

4.3.2.2. Time-varying relationship between demand factors and global crude oil prices. As shown in Fig. 4, the selected demand factors had largely accounted for more than 20 % of the total demand factors, even

reaching more than 60 % after the firstly quarter of 2020 (2020Q1), indicating that demand factors had always been the main factor influencing global crude oil prices. The demand factors have had a significant impact on global crude oil prices mainly because of five factors:

- (1) Governments of various countries took active measures to cope with the financial crisis and the world economy had gradually been recovering [49]. Prior to 2015, governments took active measures to cope with the financial crisis, while major countries such as China and the United States also strengthened their economic cooperation [39,47]. Thus, the economy of various countries gradually recovered and the demand for crude oil gradually increased, with global crude oil prices gradually rising. Therefore, the demand factor represented by economic development was the main factor driving global crude oil prices up until 2015, consistent with the findings of Kilian [5].
- (2) The external dependence of crude oil in major crude oil consuming countries such as China and India had been increasing [50]. Since the second half of 2016, with the gradual recovery of the world economy, the demand for crude oil from countries around the world, especially from large crude oil consuming countries such as China and India, had been increasing. China's external dependence on crude oil had reached 70 %, with new demand for crude oil from China's seven major refining bases exceeding 100 million tones. In 2017, non-OECD's crude oil demand, represented by China, grew by 1.23 million barrels per day to 50.45 million barrels per day and further increased by 1.35 million barrels per day to 51.8 million barrels per day in 2018. Crude oil demand continued to increase and global crude oil prices continued to rise.

The COVID-19 in 2020 had a major impact on the world economy and energy markets. The average growth rate of global GDP was -3.6 %, and there were 26 countries with a decline of more than 10 %, among which the economic growth rate of the United States was -3.5 %, and the economic growth rate of China had also declined from 6.1 % in 2019 to 2.3 %. The economic recession had greatly reduced the demand for crude oil in the world's major crude oil importers and consumers. According to the «OPEC Oil Market Monthly Report» for 2021 Issue 1st, published on January 14, 2021, the total world crude oil demand in 2020 was 900 million barrels per day, a decrease of 9.75 million barrels per day compared with 2019, a decline of 9.78 %, of which the decline in the second quarter was 17.25 %. On April 20, 2020, a historic event occurred in the global crude oil market, WTI crude oil futures settled at -\$37.63/barrel, dropping by 305.97 %, falling below 0 dollars for the

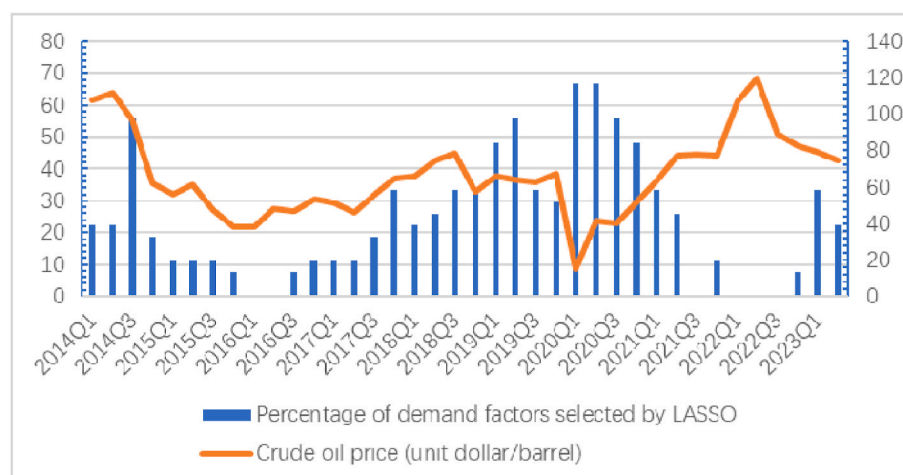


Fig. 4. The percentage of the demand factors selected by LASSO (left axis) and global crude oil prices (right axis).

first time, Brent crude oil also fell below \$9.12/barrel, dropping by 24.42 %, and the “black Swan” event appeared again. As the root cause of the recent drop in crude oil prices was the decrease in demand due to the COVID-19, the probability of the demand factor selected in 2020 is almost above 50 %, and global crude oil prices had remained at a low level.

With the strong recovery of the world economy in 2021, the demand for global crude oil was gradually increasing. The average growth rate of the global economy was 4.9 %, of which the economic growth rate of the world’s largest economy, the United States, was 5.9 %, and the economic growth rate of the second largest economy, China, was 8.1 %, and the world demand for crude oil gradually recovered. Among them, the first, second and third quarters of 2021 reached 95 %, 96 % and 97 % before the COVID-19 period, respectively, and demand recovered to 98.5 million barrels per day in December, reaching 98.3 % of the same period in 2019, 8.50 million barrels per day more than 2020, and demand factors have significantly supported for high oil prices. As a result, the probability of demand factors selected, especially in the first half of 2021, was above 20 %. However, the subsequent Russian-Ukrainian war disrupted the momentum of world economic recovery. The insecurity and economic uncertainty caused by the Russian-Ukrainian war reduced global economic growth and global crude oil demand, affecting the rise of oil prices.

In 2023, the world economy gradually got rid of the influence of the Russian-Ukrainian war, the global economic growth rate accelerated, the demand for crude oil increased, and the global crude oil prices remained high. In the first half of 2023, the global economic growth rate was 3 %, of which the US economy recovered strongly, and the economic growth rate was 2.6 %. Since the complete lifting of COVID-19 prevention and control measures by the end of 2022, Chinese economy, on the other hand, had rapidly recovered, with an economic growth rate of 5.5 % in the first half of 2023, higher than the 3.0 % for the entire year of 2022. In the first half of 2023, China imported 280 million tons of crude oil, an increase of 11.7 %. Affected by the economic recovery of major countries such as China and the United States, the global demand for crude oil was increasing, and the crude oil prices had been maintained at a high of more than 75 US dollars/barrel. Therefore, the percent of the demand factors selected in the first half of 2023 was above 20 %.

4.3.2.3. Time-varying relationship between financial factors and global crude oil prices. As shown in Fig. 5, the percentage of financial factors selected was above 30 % until the crude oil prices crash in the second half of 2014, and after the crude oil price crashed in the second half of 2014, financial factors were selected at a rate of 60 % at one point. The specific reasons were as follows:

- (i) The depreciation of U.S. dollar. The first half of 2014 saw the continued rise of the euro against the U.S. dollar, the depreciation of the U.S. dollar and the continued rise in the prices of U.S. dollar-denominated bulk commodities such as crude oil [22]. At the same time, the quantitative easing policy implemented by the Chinese central bank to ease the pressure on private borrowing further contributed to the increase in oil prices.
- (ii) Increase in base currency. Global crude oil prices continued to plummet in the second half of 2014. One of the important reasons for this was the continuous implementation of quantitative easing policies in Europe and the United States and other countries, which enhanced the liquidity of the US dollar and led to an increase in base money in the market. The large increase in base money gave speculators the opportunity to short crude oil in the futures market, thus accelerating the plunge in global crude oil prices.

While global crude oil prices were relatively flat and supply and demand factors were less correlated between 2015 and the first half of 2017, financial factors have consistently shown a strong correlation, being selected at around 10 % throughout. This was because both supply and demand for crude oil were flatter when the global crude oil prices was relatively stable, and crude oil was invested more as a risk asset with strong financial attributes. As a result, financial factors were selected.

At one point in time, the share of selected financial factors increased to 67 % as global crude oil prices surged since the second half of 2017. This was due to the improvement in the US economy, as well as the accelerated process of interest rate hikes in the US dollar as a result of Federal Reserve Chairman Powell’s preference for a tighter monetary policy, which led to a rise in the price of crude oil. The dollar-denominated nature of crude oil and the emergence of the “petrodollar” had led to speculation in crude oil as a financial asset, enhancing its financial attributes.

Under the influence of the COVID-19 in 2020, the global financial market was in turmoil, and the financial market risk began to pass to the crude oil market, causing negative global crude oil futures. On February 3, 2020, Chinese A Shanghai Stock Index plunged nearly 230 points, or 7.72 %, and the market panic soon spread to global financial markets. Speculators in the crude oil futures market withdrew a large amount of money from the crude oil futures market, resulting in a sharp fall in crude oil futures, and WTI crude oil futures once fell below \$0. At the same time, the United States had implemented large-scale fiscal stimulus and monetary easing policies in response to the economic crisis by the COVID-19, which had weakened the exchange rate of the dollar. Since international crude oil transactions were mainly priced in dollars, the depreciation of the US dollar meant that the cost of buying crude oil

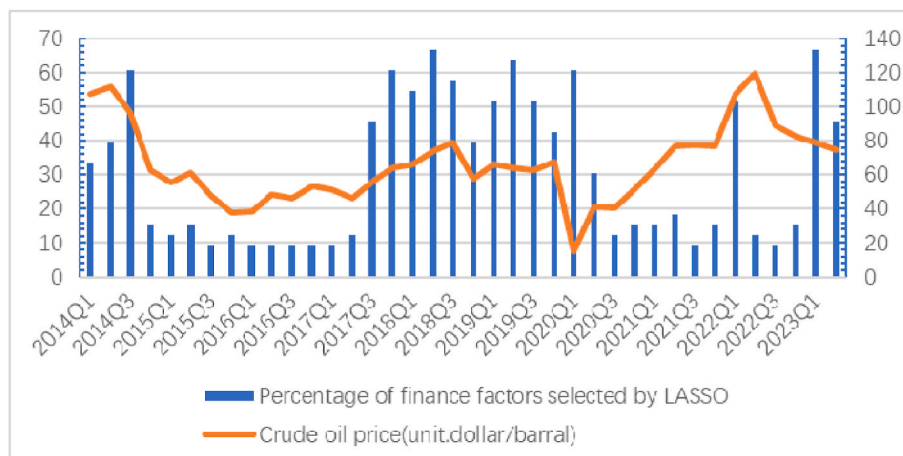


Fig. 5. The percentage of the finance factors selected by LASSO(left axis) and global crude oil prices (right axis).

in other currencies was reduced, which stimulated the rise in global crude oil prices. To sum up, financial factors such as market speculation, the spread of panic, and the depreciation of the US dollar were important factors for the plunge in global crude oil prices. As a result, the probability of the financial factor being selected at the start of COVID-19 in 2020 was more than 60 %.

In 2022, the Russia-Ukraine war caused tensions in the world economy, and the panic first fermented in the stock markets of countries such as Russia and Ukraine, and soon spread to global stock markets. Firstly, China's A Shanghai Stock Index plunged from 3481.11 points on March 3, 2022 to 3063.97 points on March 15, a drop of 417 points, or 12 %, within two weeks; After March, the Federal Reserve added statements about geopolitical risks and energy prices increases, the bullish sentiment in the crude oil futures market continued to grow, and speculative funds continued to pour into the crude oil futures market, which became an important factor to promote the rise of global crude oil prices, and the global crude oil prices soared from \$77.24/barrel in January 2022 to \$119.78 in June 2022. Therefore, in the first quarter of 2022, financial factors were selected by more than 50 % of the factors.

In 2023, the Federal Reserve opened a radical interest rate hike cycle, with 10 consecutive interest rate hikes, four of which were 75 basis points, and the US dollar's interest rate hike stimulated the strong recovery of the US economy; In addition, the hike in the dollar caused the dollar to appreciate, and the crude oil prices priced by dollars fell from \$83/barrel in the first quarter of 2023 to \$75/barrel in the second quarter of 2023.

4.3.2.4. Time-varying relationship between the midstream and downstream products of crude oil and global crude oil prices. As shown in Fig. 6, the midstream and downstream products of crude oil had a greater impact on global crude oil prices and were selected at a higher percentage, essentially above 25 %. The proportion of midstream and downstream products of crude oil remained largely at around 25 % until the third quarter of 2015 (2015Q3) and peaked in the third quarter of 2014 (2014Q3). For the reason that the midstream and downstream products of crude oil could visually reflect the price and demand of crude oil [51].

From the second half of 2017, global crude oil prices continued to rise and the proportion of the midstream and downstream products of crude oil selected increased, reaching around 50 % at one point. This was partly due to the faster global economic development, especially the increase in car ownership in large countries such as China and the United States, and the widening spread between the prices of major midstream and downstream products of crude oil, such as coal, gas and diesel, and global crude oil prices, which positively stimulated the demand for crude oil and kept its price on an increasing trend [52].

The COVID-19 in 2020 had a significant impact on global energy markets, with air transport, road transport and travel by private car

having the biggest impact. According to the International Air Transport Association (IATA) in 2021, the global air passenger traffic in 2020 would be 1.8 billion, A decrease of 60.2 % from 2019; Total passenger revenue fell 69 % year over year, resulting in a net loss of \$126.4 billion, and 66 % of the world's commercial airline fleet was grounded. In China, the world's largest oil importer, the epidemic had reduced rail passenger travel by 77 % and road passenger travel by 79 %. At the same time, the COVID-19 had also had a great influence on travel by private car. The sharp decline in air transportation, road transportation, travel, etc., had led to a decrease in the demand for middle and downstream products of crude oil, such as kerosene, gasoline, diesel, and so on, which had led to a decline in crude oil demand. Midstream and downstream products of crude oil were the direct cause of the decline in global crude oil prices. Therefore, in 2020, the ratio of midstream and downstream products of crude oil selected was basically more than 50 %, and the highest was 75 %.

In 2021, with the lifting of the ban on COVID-19 prevention and control and the gradual recovery of economy in various countries, air transport, rail transit, travel gradually recover, and the demand for middle and downstream products of crude oil also gradually recovered. Compared with 2019, the recovery degree of diesel demand in 2021 was 97 %, gasoline demand was 96 %, and naphtha demand was 106 %. Demand for jet fuel recovered by 66 %, fuel oil by 98 % and other oil by 103 % (Fig. 7). The recovery of demand for midstream and downstream products of crude oil had driven further increases in global crude oil prices, and global crude oil prices had also risen from \$51/barrel at the end of 2020 to \$77/barrel at the end of 2021. Therefore, the selection rate of middle and downstream products of crude oil, especially in the

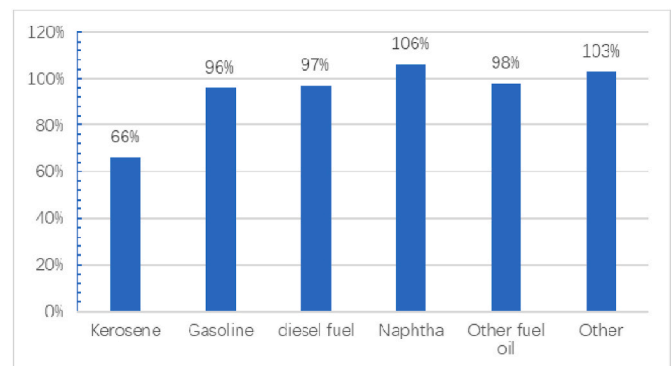


Fig. 7. Recovery degree of demand for middle and downstream crude oil products in 2021/2019
Data source:EIA.

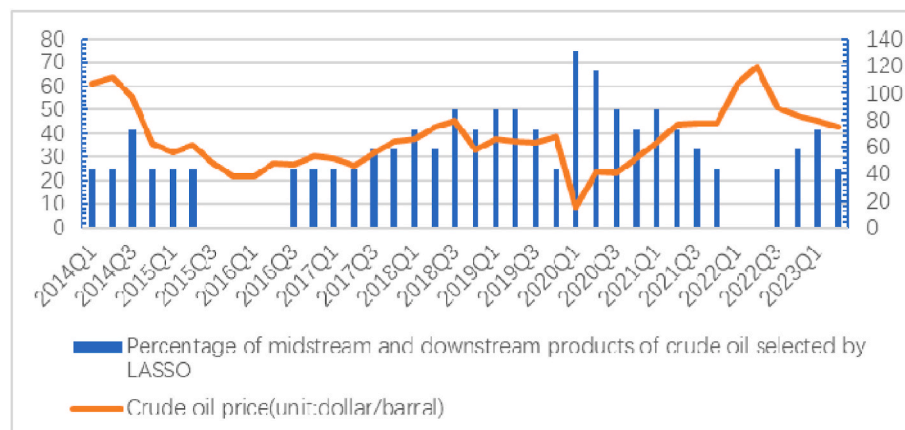


Fig. 6. The percentage of the midstream and downstream products of crude oil factors selected by LASSO (left axis) and global crude oil prices (right axis).

first half of the year, was more than 40 %.

Since 2023, with the complete liberalization of China's COVID-19 prevention and control policy, economic repair, and the continuous release of public travel demand, the overall operating condition of the air transport industry had gradually recovered. According to the report of the Civil Aviation Administration of China in 2023, in the third quarter of this year, transport aviation increased by 73.4 % year-on-year, an increase of 7.1 % over the same period in 2019; General aviation grew by 7.0 % year-on-year and 35.2 % over the same period in 2019, and other rail transit industries also gradually recovered with the full liberalization of China's COVID-19 policy. The recovery of air transportation, rail transportation and travel had increased the demand for the midstream and downstream products of crude oil such as kerosene, gasoline and diesel, which in turn had increased the demand for crude oil and effectively stopped the decline in crude oil prices.

In summary, financial factors had the greatest impact on global crude oil prices, and the midstream and downstream products of crude oil also had a very significant influence on global crude oil prices [53]. Moreover, supply and demand factors were the fundamental factors affecting global crude oil prices, with demand factors always having a greater impact on global crude oil prices, while, supply factors had a greater impact when global crude oil prices fluctuated dramatically. At the same time, sharp fluctuations in global crude oil prices were often caused by a variety of factors. For example, during the COVID-19 in 2020, the root cause of the sharp decline in global crude oil prices was the reduction in demand, and the direct cause was the withdrawal of funds caused by the reduction of kerosene, gasoline, diesel and other midstream and downstream products of crude oil and the spread of panic in the financial market. The fundamental reason for the sharp rise in global crude oil prices in 2022 was the decrease in crude oil supply, and the direct reason was the emergence of more investment in the crude oil futures market. Therefore, the sharp fluctuations in global crude oil prices in both 2020 and 2022 were caused by a combination of factors.

4.4. Structural breakpoint of global crude oil prices

From the above, it was clear that the factors that lead to crude oil prices fluctuations differed at different time periods, and in particular, the influencing factors of global crude oil prices changed more around major events. Therefore, there may be structural breakpoints in global crude oil prices [20], where the influencing factors changed significantly before and after the structural breakpoints, and the same influencing factor had different effects on global crude oil prices before and after the structural breakpoints, namely, the influencing factors of global crude oil prices were time-varying [18]. As the TVP model was able to continuously update the estimation process of the parameters and gave higher weight to the current information, the coefficients of explanatory variables changed over time. As a matter of fact, the coefficients of the TVP model changed significantly when there was a structural breakpoint in global crude oil prices. As could be seen from Section 4.3, global crude oil prices had undergone multiple major events after 2014 and significant changes in the factors influencing global crude oil prices. Therefore, the TVP model was used to estimate the coefficients on global crude oil prices and their influencing factors from the first quarter of 2014 (2014Q1) to the second quarter of 2023 (2023Q2) to capture the structural breakpoints in global crude oil prices.

The coefficients of the influencing factors estimated by the TVP model showed that the coefficients of a number of influencing factors such as the number of the global rigs, the RBOB-WTI spot prices spread, US inventories and US bond real yields all changed significantly, during the crude oil price "black swan" event in the second half of 2014, the US dollar interest rate hike in the second half of 2017, the US-China trade war in 2018 and the COVID-19 in 2020, indicating structural changes in global crude oil prices around these events. In this paper, the coefficients of two influencing factors, the number of the global rigs and US inventories, were used as examples.

As could be seen from Fig. 8, there had been four structural changes in the coefficients of the number of the global rigs and US inventories [54]. The specific analysis was as follows:

- (i). The first structural change in the price of crude oil was the "black swan" event in the second half of 2014 [55]. Around the time of the "Black Swan" event in the second half of 2014, the coefficient on the number of the global rigs went from -0.15 to 0.03 , with its impact on global crude oil prices going from negative to positive. The coefficient on US inventories also changed from -0.07 to 0.001 , with a significant change in its impact on global crude oil prices. It indicated that the "black swan" events in global crude oil prices have led to structural changes in the factors influencing global crude oil prices [56].
- (ii). The second structural change in global crude oil prices was the US dollar interest rate hike in the second half of 2017, which also led to a negative to positive coefficient change on the number of global rigs and US inventories, suggesting that the US dollar interest rate hike also led to a structural change in crude oil price influencing factors.
- (iii). The third structural breakpoints of global crude oil prices was the US-China trade war in 2018 [55,57]. caused the coefficients on these two influencing factors to change from positive to negative, suggesting that the US-China trade war also caused a structural change in the influencing factors of global crude oil prices [58].
- (iv). The fourth structural breakpoints of global crude oil prices was the COVID-19 in 2020. As COVID-19 had a significant impact on global energy markets, which had driven the coefficient of the number global rigs from -0.005 to 0.0379 . At the same time, it had also driven the US inventory coefficient from 0.211 to -0.233 . This showed that the COVID-19 had brought about structural changes in the global energy market.

To conclude, the crude oil price "black swan" event in the second half of 2014, the US dollar interest rate hike in the second half of 2017, the US-China trade war starting in 2018 and the COVID-19 in 2020 have all resulted in structural changes in the influencing factors of global crude oil prices, in line with the findings in Section 4.3. Therefore, all these four events were structural breakpoints for global crude oil prices.

5. Forecasting results based on LASSO-TVP model

As seen in above Section, different influencing factors had different impacts on global crude oil prices, with three categories of factors - finance, demand and crude oil midstream and downstream products - having a greater impact on global crude oil prices and supply factors having a smaller impact [59]. As there were noise variables and severe multicollinearity problems among these influencing factors, we used the LASSO model to perform rolling screening of these influencing factors for the very first.

5.1. Key factors selected by Lasso model

As shown in Table 4, when there were no the midstream and downstream products of crude oil, the LASSO model screened 25 influencing factors with a frequency of more than 10 %, accounting for 86.2 % of all factors, however there were only 18 influencing factors when there were the midstream and downstream products of crude oil, accounting for only 54.5 % of all influencing factors. This showed that the information contained in the midstream and downstream products of crude oil was difficult to replace. Besides, Naphtha in the crude oil midstream and downstream product was selected more frequently at 89 times, which was selected close to 80 %. RBOB-WTI spot prices spread was selected the most, which was also closely related to the midstream and downstream products of crude oil. So that the midstream and downstream products of crude oil were very important to the crude oil

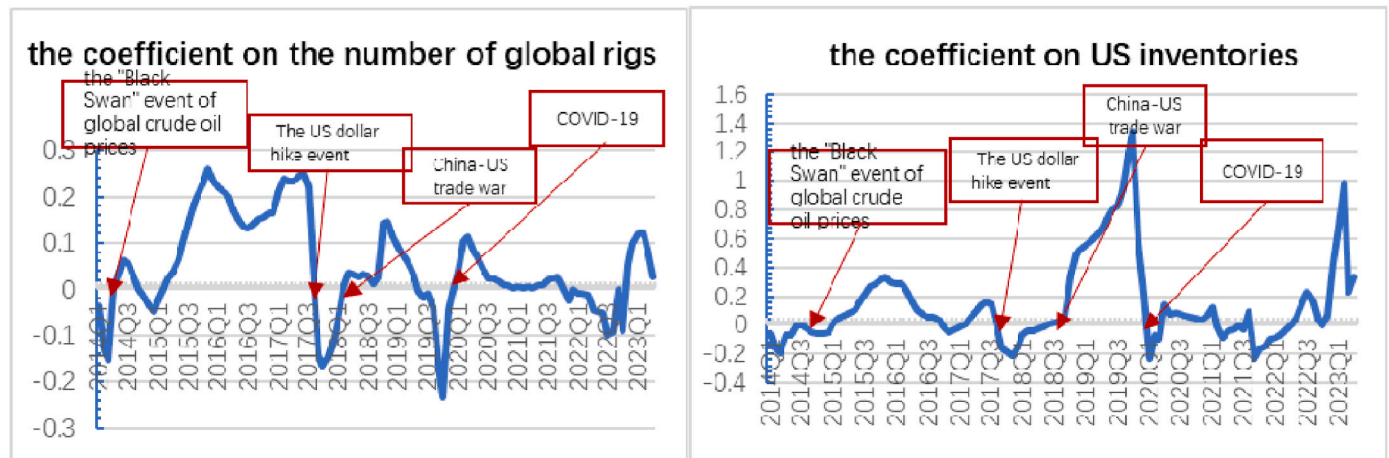


Fig. 8. The coefficient of the number of global rigs estimated by the TVP model (left map) and the coefficient of the United States by the TVP model (right map).

Table 4

The influencing factors which's frequency by LASSO exceeds 10 %.

the frequency of the influencing factors selected by LASSO exceeds 10 % without the midstream and downstream products of crude oil			the frequency of the influencing factors selected by LASSO exceeds 10 % with the midstream and downstream products of crude oil		
Variables	Selected frequency	Selected percentage (%)	Variables	Selected frequency	Selected percentage (%)
RBOB-WTI spot price spread	98	86.7	RBOB-WTI spot price spread	94	83.2
US bond real yield	74	65.5	Naphtha ¥/ton	89	78.8
Inflation rate	69	61.1	US bond real yield	80	70.8
Credit risk premium	57	50.4	Credit risk premium	68	60.2
Number of American rigs	55	48.7	Inflation rate	66	58.4
VIX	55	48.7	US inventory	59	52.2
LIBOR	53	46.9	American crude oil export volume	59	52.2
US dollar index	51	45.1	Number of global rigs	56	49.6
Chinese economic growth	46	40.7	LIBOR	56	49.6
CRB index	44	38.9	VIX	55	48.7
OPEC residual capacity	42	37.2	US dollar index	46	40.7
US inventory	40	35.4	WTI futures position	42	37.2
Chinese crude oil import and processing volume	39	34.5	HH Natural gas price	42	37.2
WTI futures position	39	34.5	US shale oil production	40	35.4
US shale oil production	38	33.6	CRB index	40	35.4
Global crude oil supply	35	31.0	Kerosene ¥/ton	30	26.5
HH Natural gas price	29	25.7	Gold price	20	17.7
American refining production	28	24.8	China PMI	18	15.9
US PMI	23	20.4			
S&P 500 Index	21	18.6			
China PMI	21	18.6			
Number of American rigs	17	15.0			
Gold price	13	11.5			
American crude oil production	13	11.5			
Total:25			Total:18		

prices.

5.2. Forecasting results of LASSO-TVP model with midstream and downstream products of crude oil

Since LASSO screened out factors with greater influence and excluded factors with less influence, the influencing factors screened out based on the LASSO model had a strong influence [22]. Since the midstream and downstream of crude oil products had a strong influence on global crude oil prices, we substituted the 18 influencing factors with a frequency of more than 10 %, which are screened by LASSO when the midstream and downstream products of crude oil were available, into the TVP model to construct the LASSO-TVP model to forecast global crude oil prices and compare the forecasting results with those of other models, so as to verify whether the LASSO-TVP model had a strong

Table 5

Prediction results of each model.

one-step forecasting	LASSO-TVP	LASSO	TVP	No-change
MAPE	3.6 %(1)	9.8 %(2)	25.3 %(4)	15.3 %(3)
MAE	0.125(1)	0.322(2)	0.890(4)	0.491(3)
two-step forecasting	LASSO-TVP	LASSO	TVP	No-change
MAPE	4.1 %(1)	11.7 %(2)	36.6 %(4)	16.9 %(3)
MAE	0.133(1)	0.441(2)	1.117(4)	0.492(3)
three-step forecasting	LASSO-TVP	LASSO	TVP	No-change
MAPE	4.4 %(1)	12.2 %(2)	41.2 %(4)	16.5 %(3)
MAE	0.157(1)	0.458(2)	1.335(4)	0.598(3)
fourth-step forecasting	LASSO-TVP	LASSO	TVP	No-change
MAPE	5.1 %(1)	14.9 %(2)	57.5 %(4)	18.2 %(3)
MAE	0.196(1)	0.566(2)	2.028(4)	0.703(3)

Note: Number in parentheses denotes ranking of monthly minimum error values.

forecasting ability when the midstream and downstream products of crude oil were available. The results were shown in Table 5.

Table 5 showed the error values for the four forecasting methods, LASSO-TVP model, LASSO model, TVP model and No-change model, at 1, 2, 3 and 4 steps of forecast length under a 9-year fixed window. As shown in Table 5, the error results of the four models were analyzed as follows:

- (i). LASSO-TVP model had the best prediction effect. As shown in Table 5, among the one-step, two-step, three-step and four-step predictions, the prediction error of LASSO-TVP model was the smallest and far smaller than that of other models. Taking the MAPE of one-step prediction as an example, the MAPE of the LASSO-TVP model was 3.6 %, much smaller than the 25.3 % of the TVP model and the 9.8 % of the LASSO model, which was the smallest of the four models. Therefore, the prediction effect of LASSO-TVP model was the best. This was because the LASSO model could screen out the factors with the strongest prediction effect and eliminate the factors with poor prediction effect, which reduced the complexity of the model and the noise of the data. The TVP model could not only update the data in time, but also identify the structural breakpoint of the crude oil prices. Therefore, compared with the LASSO model, TVP model and No-change model, the LASSO-TVP model could predict the global crude oil prices more accurately.
- (ii). The stronger the correlation of the input data of the model, the less data, the better the prediction effect. As can be seen from Table 5, the prediction effect of LASSO model was better than that of TVP model. This was because the LASSO model selected 18 highly correlated data for prediction, which was half of the input layer of the TVP model, which not only selected the strongest factors for prediction effect, but also reduced the model complexity and noise, so the prediction effect was better than the TVP model. However, the prediction effect of LASSO model was smaller than that of LASSO-TVP model because it couldn't update the prediction information in time and identified the structural breakpoint. Although the TVP model could update the data in time and identify structural breakpoints, it couldn't screen out the data with the best prediction effect, resulting in too many input data (33 in total), and many data with weak correlation and too much noise, so the prediction effect was the worst. The reason why the No-change model could be better than the TVP model was that after 2019, the global crude oil prices fluctuated greatly, the prediction was difficult, and many models cannot accurately predict, and the No-change model had certain advantages over the TVP model.
- (iii). The predictive error increased gradually with the increase of forecasting steps. As shown in Table 5, among the four models, one-step prediction had the best effect and four-step prediction had the worst effect. This was because the length of the one-step prediction was shorter and the information was newer, while the four-step prediction was longer and the information was older, so the newer the prediction information was, the better the prediction effect was.

In conclusion, the LASSO-TVP model had the best prediction effect, because the LASSO-TVP model could not only screen out the factors with strong prediction effect, but also updated the prediction information in time and identified the structural breakpoints. It combined the advantages of LASSO and TVP models, so the prediction effect was better. At the same time, the newer the prediction information, the better the prediction effect.

5.3. Forecasting results of LASSO-TVP model without midstream and downstream products of crude oil

As could be seen from Section 5.1, the midstream and downstream products of crude oil were selected by the LASSO model with high frequency and had a greater impact on global crude oil prices, and it could provide better forecast of global crude oil prices by the midstream and downstream products of crude oil. As the use of the midstream and downstream products of crude oil in forecasting global crude oil prices had been rare in the past, so in this paper, the four models were constructed to forecast global crude oil prices by the influencing factors without the midstream and downstream products of crude oil, and comparing the forecasting results of the four models to test whether the LASSO-TVP model had the strongest forecasting ability.

Table 6 represented the predictive errors for steps 1, 2, 3 and 4 of the four models without the midstream and downstream products of crude oil. The results were as follows:

- (i) The LASSO-TVP model had the best forecasting performance. Among the four prediction steps, the prediction error of LASSO-TVP model was the smallest. Taking the MAPE of one-step prediction as an example, the MAPE of the LASSO-TVP model was 9.8 %, which was smaller than 13.2 % of the LASSO model, 15.3 % of the No-change model and 17.5 % of the TVP model, and the error was the smallest among the four models. This indicated that the prediction effect of LASSO-TVP model was robust, because it combined the advantages of LASSO model and TVP model, which could not only screen out the factors with strong prediction effect, but also updated the prediction information in time. Therefore, the prediction effect of LASSO-TVP model was the best.
- (ii). The midstream and downstream products of crude oil played a very important role in the prediction of crude oil price. As can be seen from Table 6, the prediction error of the four models was larger and the prediction effect was worse when there were no midstream and downstream products of crude oil. This was because the midstream and downstream products of crude oil had a very important impact on the global crude oil prices, and other factors were difficult to replace. The LASSO-TVP model, LASSO model, and TVP model all became worse when there were no midstream and downstream products of crude oil. Therefore, the midstream and downstream products of crude oil had more forecasting information, which was helpful to improve the forecasting effect of crude oil prices.

5.4. DM test

To validate the superiority of the LASSO-TVP model, we used the DM test. The test results of DM statistics with their p values were shown in Table 7 for both the midstream and downstream products of crude oil and no the midstream and downstream products of crude oil [60,61].

Table 6

Prediction results of each model without the midstream and downstream products of crude oil.

one-step forecasting	LASSO-TVP	LASSO	TVP	No-change
MAPE	9.8 %(1)	13.2 %(2)	17.5 %(4)	15.3 %(3)
MAE	0.355(1)	0.461(2)	0.611(4)	0.491(3)
two-step forecasting	LASSO-TVP	LASSO	TVP	No-change
MAPE	10.4 %(1)	15.9 %(2)	19.1 %(4)	16.9 %(3)
MAE	0.391(1)	0.557(2)	0.693(4)	0.492(3)
three-step forecasting	LASSO-TVP	LASSO	TVP	No-change
MAPE	11.7 %(1)	16.3 %(2)	20.8 %(4)	16.5 %(3)
MAE	0.409(1)	0.594(2)	0.797(4)	0.598(3)
fourth-step forecasting	LASSO-TVP	LASSO	TVP	No-change
MAPE	12.1 %(1)	17.7 %(2)	35.7 %(4)	18.2 %(3)
MAE	0.447(1)	0.606(2)	1.120(4)	0.703(3)

Note: Number in parentheses denotes ranking of monthly minimum error values.

Table 7
DM test results.

Model	LASSO	TVP	No-change
the midstream and downstream products of crude oil			
LASSO-TVP	−4.72 (0.00) ^a	−8.01 (0.00) ^a	−16.71 (0.00) ^a
without the midstream and downstream products of crude oil			
LASSO-TVP	−6.69 (0.003) ^a	−10.25 (0.00) ^a	−16.06 (0.00) ^a

^a Significant at 1 %.

As shown in Table 7, the DM test results statistically confirmed the above conclusion. The proposed model LASSO-TVP model statistically outperformed LASSO, TVP and No-change models with their respective p values, and their p values were far below than 0.01 for both the midstream and downstream products of crude oil and no the midstream and downstream products of crude oil.

6. Discussion

This paper analyzed the impact of influencing factors on global crude oil prices from five perspectives: supply, demand, finance, alternative product and the midstream and downstream products of crude oil, screened out the factors with greater impact by the LASSO model and analyzes them. In addition, the TVP model was used to identify the structural breakpoint of global crude oil prices [62], and finally used the LASSO-TVP model to forecast global crude oil prices and drew the following conclusions:

- (i). In addition to supply and demand factors, financial factors and alternative product factors, the midstream and downstream products of crude oil had important influence on global crude oil prices. It could be seen from Section 4.3.2.4 that the midstream and downstream products of crude oil had a significant impact on crude oil prices at different stages. Especially during the COVID-19 in 2020, air transportation, rail transportation, travel, etc., were greatly reduced, which resulted in a sharp reduction of the midstream and downstream products of crude oil, such as kerosene, gasoline, and diesel, and finally led to a sharp decline in global crude oil prices. Therefore, during the COVID-19, the midstream and downstream products of crude oil were the most direct cause of the plunge in global crude oil prices. It could be seen from Section 5.1 that the midstream and downstream products of crude oil had an important impact on global crude oil prices and contained a lot of information about crude oil price prediction. Therefore, the midstream and downstream products of crude oil had the highest probability of being selected and were difficult to be replaced. Therefore, in Section 5.2 and 5.3, the prediction effect of the midstream and downstream products of crude oil was better than that without the midstream and downstream products of crude oil. When choosing the factors of global crude oil prices, the midstream and downstream products of crude oil should not be ignored.
- (ii). The LASSO model was able to filter out factors with greater influence and better predictive power. As could be seen from Section 4, the influencing factors screened out by LASSO model were all factors that were closely related to global crude oil prices, and the factors screened out by LASSO at different stages were the main causes of crude oil prices fluctuations. At the same time, the influencing factors screened by the LASSO model also had good forecasting ability.
- (iii). The influencing factors of global crude oil prices differed in different time periods, and the same influencing factor had different impacts on global crude oil prices in different time periods. For example, the plunge of global crude oil prices in the second half of 2014 was mainly caused by supply factors such as

the increase in shale oil production and the restoration of production capacity by traditional oil-producing countries, while the sharp rise of global crude oil prices in 2017 was due to the gradual economic recovery of countries such as China and India, the increase in crude oil external dependence and the interest rate hike of the US dollar [63,64].

- (iv). There was structural movement in global crude oil prices, with different influencing factors before and after the structural break point. According to the results of the TVP model, the crude oil price “black swan” event in 2014, the US dollar interest rate hike in the second half of 2017, the US-China trade war in 2018 and the COVID-19 in 2020 were all structural breakpoints for global crude oil prices, and the influencing factors of global crude oil prices and their coefficients, varied significantly before and after them.
- (v). The LASSO-TVP model combined the advantages of LASSO model and TVP model, which could screen out the factors that have a greater impact on global crude oil prices, and update the crude oil price prediction information in time, so it could better predict the global crude oil prices. The results in Tables 4 and 5 and the DM test had been demonstrated.
- (vi). Strong fluctuations of global crude oil prices were often caused by a combination of factors [16,65]. For example, in Section 4.3, the collapse of crude oil prices due to the COVID-19 in 2020 was caused by the combination of middle and downstream products of crude oil, crude oil demand factors, and financial factors, while the surge in crude oil prices in 2022 was caused by supply factors and financial factors. Among them, supply and demand factors were the fundamental reasons leading to sharp fluctuations in crude oil prices, while financial factors and midstream and downstream products were the direct reasons leading to sharp fluctuations in global crude oil prices.

7. Conclusion

This paper selected 33 influencing factors of global crude oil prices from five perspectives, and took the LASSO model to screen and analyze them. From the discussion in Section 6, it could be seen that midstream and downstream products of crude oil had an important impact on crude oil prices, and the sharp fluctuation of global crude oil prices was caused by multiple factors. The influencing factors of global crude oil prices were different in different time periods, and the same influencing factors had different effects on global crude oil prices in different time periods. In addition, there were structural changes in global crude oil prices, and the influencing factors of global crude oil prices were different before and after the structural break point. At the same time, the prediction effect of LASSO-TVP model was better than that of LASSO model and TVP model. In conclusion, we made the following suggestions:

- (1) the midstream and downstream products of crude oil contained strong predictive information, so the midstream and downstream products of crude oil should be taken into account when predicting the price of crude oil. For example, during the COVID-19 in 2020, the midstream and downstream products of crude oil had a huge impact on crude oil prices, which was the direct cause of the plunge in crude oil prices.
- (2) when the global crude oil prices fluctuated violently, we should consider a number of influencing factors to forecast the global crude oil prices. In the long run, supply and demand factors were the root cause of global crude oil prices fluctuation, financial factor and midstream and downstream products of crude oil were the direct cause of global crude oil prices fluctuation.
- (3) structural breakpoints should be considered when forecasting global crude oil prices. Because the structural break point of global crude oil prices was abnormal fluctuation, some scholars remove the structural break point of global crude oil prices before making the forecast. Therefore, we suggested that we can remove the structural breakpoint and then forecast the global crude oil prices in the future.
- (4) since the LASSO-TVP model had achieved a good

prediction effect in predicting global crude oil prices, it showed that the mixed model combines the advantages of the single model. Therefore, we suggested to take the mixed model to forecast global crude oil prices in the future.

There were several limitations and directions that we could extend our work: (1) Geopolitics was also an important factor of global crude oil prices. Because geopolitics was difficult to quantify, this paper did not consider geopolitical factors. We would quantify and apply geopolitics to the prediction of global crude oil prices in the future. (2) There were many prediction models for crude oil prices, and many scholars used LSTM model, SVM model and ARIMA model, which are not used in this paper. Therefore, we would take neural networks, artificial intelligence and hybrid models to forecast crude oil prices. (3) As the industry was more interested in the direction fluctuations of crude oil prices, this paper did not predict the direction of global crude oil prices. Therefore, in the future, we would forecast the direction of global crude oil prices to provide reference for the industry.

CRedit authorship contribution statement

Donghua Wang: Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing – review & editing, Project administration. **Tianhui Fang:** Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft.

Ethics approval and consent to participate

This article does not involve the ethical problem.

Consent to publication

Not applicable.

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Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position in, or the review of, the manuscript entitled.

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Data availability

The authors do not have permission to share data.

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